

Question Bank for MSE FOR DTSP SUBJECT

Chapter 1-DFT and FFT

1. Compute the DFT of the 4-pt sequence $x(n) = \{0, 1, 2, 3\}$

2. Compute the sequence $x(n)$ for which the DFT is $\{20, -4+4j, -4, -4-4j\}$

3. For a discrete time sequence $x(n) = \{1, 2, 3, 4\}$, DFT is given by

 $X(k) = \{10, -2+2j, -2, -2-2j\}$. Compute the DFT of $x(n) = \{3, 4, 1, 2\}$ using the circular time shift property of DFT.

4. For an 8-point DFT, the first five DFT coefficients are

$X(k) = \{10, 2+3j, 1+2j, j, 4\}$. Find the remaining coefficients.

5. Perform circular convolution on the two sequences $x_1(n) = \{1, 2, 3, 4\}$, $x_2(n) = \{4, 1, 1, 2\}$ using Matrix multiplication method.

6. If $x(n) = \{1, 2, 3, 4, 5, 6, 7\}$ and $h(n) = \{1, 0, 2\}$. Perform convolution using the Overlap-Add method.

7. Find the Linear convolution using the Overlap-save method of the following sequences:

$x_1(n) = \{1, 2, -1, 2, 3, -2, -3, -1, 1, 1, 2, -1\}$

$h(n) = \{1, 2, 3\}$

8. Find 8-pt DFT of the sequence $x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$ using Radix-2 DIT-FFT.

9. Find 8-pt DFT of the sequence $x(n) = \{-1, 0, 2, 0, -4, 0, 2, 0\}$ using Radix-2 DIF-FFT.

10. How many computations are required to compute a 16-point DFT using the DFT and FFT algorithms?

11. Compute 4-pt DFT of a sequence $x(n)=\{0,1,2,3\}$ using DIT-FFT and DIF-FFT.

Chapter 2- IIR Digital Filter

1. Write a short note on the characteristics of the Butterworth filter approximation.

2. Write a short note on the characteristics of the Chebyshev filter (Type 1 and Type 2) approximation.

3. Show the mapping from s-plane to z-plane using Impulse invariance method.

4. Explain filter classification based on pass-band for Low Pass Filter (LPF), High Pass Filter (HPF), Band Pass Filter (BPF), Band Stop Filter (BSP).

5. Obtain a digital filter transfer function (TF) by applying impulse invariance transformation on the analog TF.

$H_a(s)=s/s^2+3s+2$. Use $F_s=1K$ samples/sec

6. Difference between the Impulse Invariance Method and the Bilinear Transformation Method.

7. The transfer function of analog filter is $H(s)=3/(s+2)(s+3)$ with $T_s=0.1$ sec. Design the digital IIR filter using BLT.

8. Using bilinear transformation, design a Butterworth filter which satisfies the following condition:

$$0.8 < |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2, 0.6\pi \leq \omega \leq \pi$$

9. Design a low-pass 1 rad/sec bandwidth Chebyshev filter with the following characteristics.

- (i) Acceptable pass band ripple of 2 dB.
- (ii) Cut-off radian frequency of 1 rad/sec.
- (iii) Stop band attenuation of 20 dB or greater beyond 1.3 rad/sec.

Chapter 3 -FIR Digital Filters

1.Explain the characteristics of FIR Filters.

2.FIR filter have impulse responses as given below. Classify them as minimum phase, maximum phase, or mixed phase.

(i) $h_1(n) = \{1, 0.707, 0.25\}$

(ii) $h_2(n) = \{1, 1.414, 1\}$

(iii) $h_3(n) = \{1, -5, 6\}$

3. Why are FIR filters called linear phase FIR filters?

4. Differentiate between FIR and IIR Digital filters.

5.Design a FIR bandpass filter to meet following specifications using frequency sampling method.

(i) Cut-off frequency=1000 Hz and 3000 Hz.

(ii) Sampling frequency=8000 Hz

(iii) Length of the filter $N=7$

OR

Using the frequency sampling method, design a low-pass FIR filter to meet the following specifications:

Passband:0-5KHz

Sampling frequency:18 KHz

Filter length=9

6. Design a linear phase FIR lowpass filter of length 7 with cut-off frequency of 1 rad/sec using a rectangular window.

7. Design a digital FIR filter for the following specifications.

Use a Hamming window and assume $M=7$.

